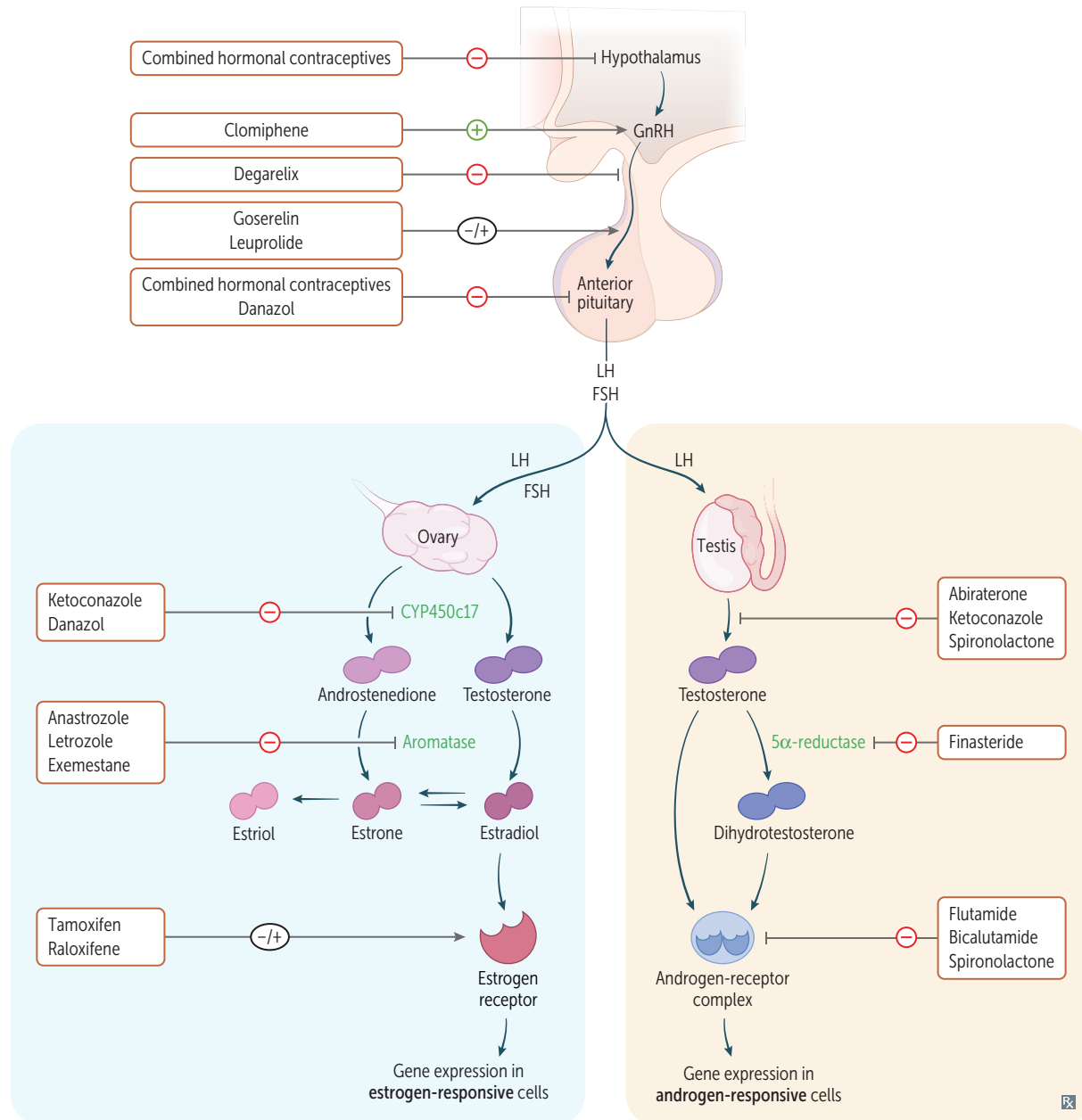


► REPRODUCTIVE—PHARMACOLOGY

Control of reproductive hormones



Gonadotropin-releasing hormone analogs**Leu**prolide, goserelin, nafarelin, histrelin.

MECHANISM	Act as GnRH agonists when used in pulsatile fashion. When used in continuous fashion, first transiently act as GnRH agonists (tumor flare), but subsequently act as GnRH antagonists (downregulate GnRH receptor in pituitary → ↓ FSH and ↓ LH → ↓ estrogen in females and ↓ testosterone in males). Can be used in lieu of GnRH.
CLINICAL USE	Uterine fibroids, endometriosis, precocious puberty, prostate cancer, infertility. P ulsatile for p regnancy, c ontinuous for c ancer.
ADVERSE EFFECTS	Hypogonadism, ↓ libido, erectile dysfunction, nausea, vomiting.

Degarelix

MECHANISM	GnRH antagonist. No start-up flare.
CLINICAL USE	Prostate cancer.
ADVERSE EFFECTS	Hot flashes, liver toxicity.

Ethinyl estradiol

MECHANISM	Binds estrogen receptors.
CLINICAL USE	Hypogonadism or ovarian failure, contraception (combined with progestins), HRT in postmenopausal females (use for < 5 years to lower risk of adverse effects).
ADVERSE EFFECTS	↑ risk of endometrial cancer (when given without progesterone), bleeding in postmenopausal patients, hepatic adenoma, breast tenderness, ↑ risk of thrombi. Contraindications—people > 35 years old who smoke tobacco (↑ risk of cardiovascular events), patients with ↑ risk of cardiovascular disease (including history of venous thromboembolism, coronary artery disease, stroke), migraine (especially with aura), breast cancer, liver disease.

Selective estrogen receptor modulators

Clomiphene	Antagonist at estrogen receptors in hypothalamus. Prevents normal feedback inhibition and ↑ release of LH and FSH from pituitary, which stimulates ovulation. Used to treat infertility due to anovulation (eg, PCOS). May cause hot flashes, ovarian enlargement, multiple simultaneous pregnancies, visual disturbances.
Tamoxifen	Antagonist at breast, partial agonist at uterus, bone. Hot flashes, ↑ risk of thromboembolic events (especially with tobacco smoking), and endometrial hyperplasia and/or cancer. Used to treat and prevent recurrence of ER/PR ⊕ breast cancer and to prevent gynecomastia in patients undergoing prostate cancer therapy.
Raloxifene	Antagonist at breast, uterus; agonist at bone; hot flashes, ↑ risk of thromboembolic events (especially with tobacco smoking), but no increased risk of endometrial cancer (vs tamoxifen, so you can “ relax ”); used primarily to treat postmenopausal osteoporosis.

Aromatase inhibitors Anastrozole, letrozole, exemestane.

MECHANISM Inhibit peripheral conversion of androgens to estrogen.

CLINICAL USE ER ⊕ breast cancer in postmenopausal females. Letrozole can be used to treat infertility from anovulation (eg, PCOS).

Progestins Levonorgestrel, medroxyprogesterone, etonogestrel, norethindrone, megestrol.

MECHANISM Bind progesterone receptors, ↓ growth and ↑ vascularization of endometrium, thicken cervical mucus.

CLINICAL USE Contraception (forms include pill, intrauterine device, implant, depot injection), endometrial cancer, abnormal uterine bleeding. Progestin challenge: presence of bleeding upon withdrawal of progestins excludes anatomic defects (eg, Asherman syndrome) and chronic anovulation without estrogen.

Antiprogestins Mifepristone, ulipristal.

MECHANISM Competitive inhibitors of progestins at progesterone receptors.

CLINICAL USE Termination of pregnancy (mifepristone with misoprostol); emergency contraception (ulipristal).

Contraception Birth control.

	METHOD	MECHANISM	NOTES
Hormonal	Estrogen combined with progestins	Prevent ovulation by ↓ GnRH → ↓ LH/FSH → no estrogen surge → no LH surge	Forms include pill (OCPs), transdermal patch, vaginal ring
	Progestin-only	Progestins also thicken cervical mucus (↓ sperm entry) and thin endometrium (less suitable for implantation)	
Intrauterine device	Copper IUD (hormone free)	Copper IUD causes local inflammation that is toxic to sperm and ova preventing fertilization and implantation	IUDs ↑ risk for abnormal uterine bleeding; insertion contraindicated in patients with active STI and heavy menstrual bleeding
	Progesterone IUD	Same as progestins	
Surgical	Males—vasectomy	No sperm in ejaculate	Irreversible
	Females—tubal ligation	Sperm cannot reach ova	

Tocolytics

Medications that relax the uterus; include **T**erbutaline (β₂-agonist action), **I**ndomethacin (NSAID), **N**ifedipine (Ca²⁺ channel blocker). Used to ↓ contraction frequency in preterm labor and allow time for administration of glucocorticoids (to promote fetal lung maturity) or transfer to appropriate medical center with obstetrical care. Keep baby in the **TIN**.

Danazol

MECHANISM	Synthetic androgen that acts as partial agonist at androgen receptors.
CLINICAL USE	Endometriosis, hereditary angioedema.
ADVERSE EFFECTS	Weight gain, edema, acne, hirsutism, masculinization, ↓ HDL levels, hepatotoxicity, idiopathic intracranial hypertension.

Testosterone, methyltestosterone

MECHANISM	Agonists at androgen receptors.
CLINICAL USE	Treat hypogonadism and promote development of 2° sex characteristics.
ADVERSE EFFECTS	Virilization in females; testicular atrophy in males. Premature closure of epiphyseal plates. Risk of acne, ↑ erythrocytosis, ↑ LDL, ↑ HDL.

Antiandrogens

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Abiraterone	17 α -hydroxylase/17,20-lyase inhibitor (↓ steroid synthesis)	Prostate cancer	Hypertension, hypokalemia (↑ mineralocorticoids)
Finasteride	5 α -reductase inhibitor (↓ conversion of testosterone to DHT)	BPH, androgenetic alopecia	Gynecomastia, sexual dysfunction
Flutamide, bicalutamide	Nonsteroidal competitive inhibitors at androgen receptor (↓ steroid binding)	Prostate cancer	Gynecomastia, sexual dysfunction
Ketoconazole	17 α -hydroxylase/17,20-lyase inhibitor	Prostate cancer	Gynecomastia
Spirolactone	Androgen receptor and 17 α -hydroxylase/17,20-lyase inhibitor	PCOS	Amenorrhea

Tamsulosin

MECHANISM	α_1 -antagonist selective for $\alpha_{1A/D}$ receptors in prostate (vs vascular α_{1B} receptors) → ↓ smooth muscle tone → ↑ urine flow.
CLINICAL USE	BPH.

Minoxidil

MECHANISM	Direct arteriolar vasodilator.
CLINICAL USE	Androgenetic alopecia, severe refractory hypertension.